EXPERIMENT-07:

Write a program to implement Logistic Regression (LR) algorithm in python

CODE:

```
# Import required libraries
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score, classification_report

# Load the dataset
data = load_breast_cancer()
X = data.data
y = data.target

# Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y
)

# Create the Logistic Regression model
model = LogisticRegression(max_iter=5000)
```

```
# Train the model
model.fit(X_train, y_train)

# Predict the test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy * 100, "%")

# Display confusion matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)

# Plot confusion matrix
plt.figure(figsize=(6, 4))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=data.target_names,
    yticklabels=data.target_names
)
```

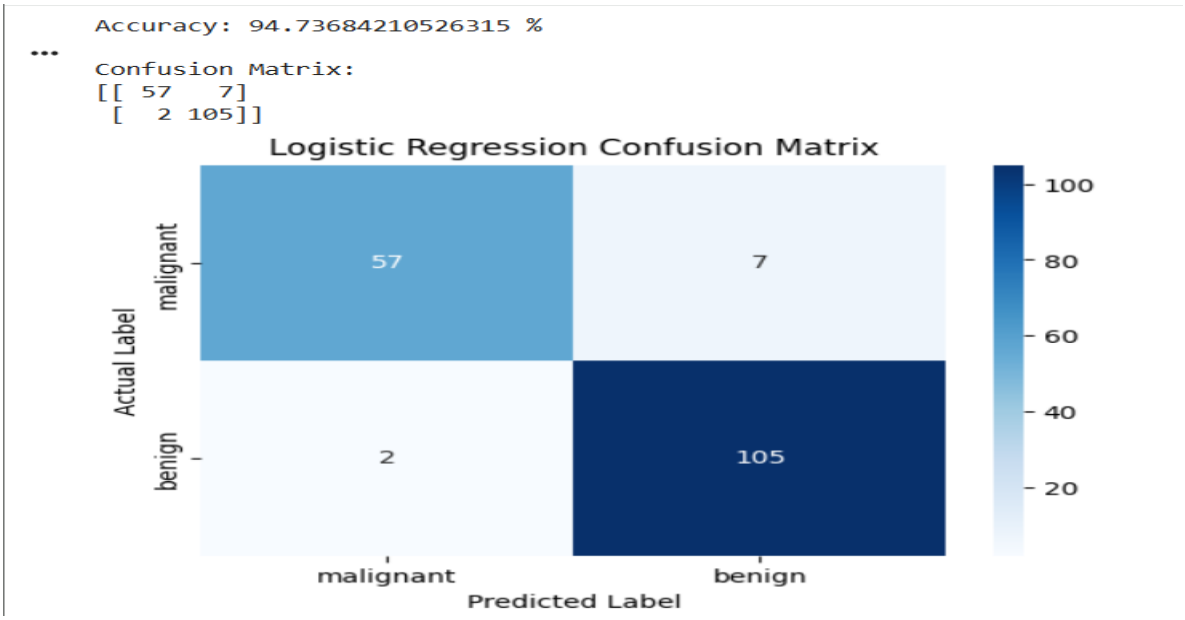
```
plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.title("Logistic Regression Confusion Matrix")
plt.show()

# Display classification report
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=data.target_names))
```

OUTPUT:

```
Accuracy: 94.73684210526315 %

Confusion Matrix:
[[ 57   7]
 [  2 105]]
```



Classification Report:

	precision	recall	f1-score	support
malignant	0.97	0.89	0.93	64
benign	0.94	0.98	0.96	107
accuracy			0.95	171
macro avg	0.95	0.94	0.94	171
weighted avg	0.95	0.95	0.95	171